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Revision History		
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1	-	On the basis of the Change-PLN-Corp-16920, this procedure has been released

1 Purpose

This procedure defines the process for servicing of distributed finished devices manufactured by Luminex Corporation.

2 Applicability

This procedure applies to Luminex sites and Diasorin subsidiaries (with the exclusion of Brazil, Czech Republic, India, Israel and Mexico) related to the Service activities for Luminex products distributed to customers and returned to Luminex Corporation.

3 Applicable Documents (external origin)

- The following list represents the external documents that apply to the present procedure; applicable revisions are listed in the Corp-Attach-005098.
 - ISO 9001 - Quality Management System - Requirements.
 - ISO 13485 - Medical Devices. Quality Management Systems. Requirements for Regulatory Purposes.
 - Directive 98/79/EC of the European Parliament and of the Council of 27 October 1998.
 - FDA 21 CFR Part 11, Electronic Record- Electronic Signatures.
 - FDA 21 CFR Part 820, Quality System Regulation
 - EU Regulation 2017/746 on in vitro diagnostic medical devices
- Documents listed in the table here below are available in Luminex Quality Management System:

Luminex applicable Document #	Document Title
QMS-021	Statistical Techniques
00091	Training Operating Procedure
00093	Inspection, Measuring and Test Equipment Standard Operating Procedure
00166	Global Support Services Communication Operating Procedure
00283	Corrective and Preventive Action Standard Operating Procedure
00370	Product Recall & Field Actions Standard Operating Procedure
00378	Legal Contract Standard Operating Procedure
00935	Returning Defective Inventory To Manufacturing Standard Operating Procedure
00942	Inquiry and Complaint Handling Operating Procedure
01199	Field Service Parts Return and Management Standard Operating Procedure
01651	Field Service Technical Documentation Standard Operating Procedure
01836	Supplier Qualification and Management Program Standard Operating Procedure
02151	Building Manuals and Package Inserts Standard Operating Procedure

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Luminex applicable Document #	Document Title
02381	Corporate Distribution Programs Operating Procedure
02487	Risk Management Standard Operating Procedure
02643	Unique Device Identification (UDI) Standard Operating Procedure
02882	QMS Software Validation Standard Operating Procedure
02883	QMS Software Master List
03003	Calibration and Preventative Maintenance OP
03069	Failure Modes and Effects Analysis (FMEA) Standard Operating Procedure
03491	Global Service Operating Procedure
03670	Returned Materials Authorization Standard Operating Procedure
03719	Routine Servicing List
03735	New Field Service Engineer Onboarding Point of Use Instructions
03756	Work Order Documentation Point of Use Instructions
03784	MDx and LTG Work Order Review and Closure Point of Use Instructions
03805	Field Quality Work Order Assessment Point of Use Instructions
03971	Field Service Inventory Management SOP
03999	Parts Ordering Point of Use Instructions
04080	Field Service Work Order Management SOP
04437	Creating Field Service Product Training Certification SOP
04445	Field Service Training Content Development SOP

4 Definitions

Term	Definition
Device Identifier (DI)	Refer to 02643
Field Service Manual	A document used by Field Service Engineers to perform repair and replace procedures.

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Correction	Repair, modification, adjustment, re-labeling, destruction, or inspection (including patient monitoring) of a device without its physical removal from its point of use to some other location.
Complaint	Any written, electronic, or oral communication that alleges deficiencies related to the identity, quality, durability, reliability, safety, effectiveness, or performance of a medical device that has been released from the organization's control or related to a service that affects the performance of such medical device. The complaint is the demonstration of dissatisfaction reported by a customer about requirements of products and their distribution, which the customer expects an answer and solution for. Product complaints are specifically linked to quality and/or performance of a product/component, as it is placed on the market (labelling and packaging included). e.g.: - Performance inferior to the claimed one as defined by product release specifications; - Mislabeled products; - Bacteriological, chemical, physical change or deterioration of the product; - Reaction or injury to a person related to the use of the product, whether or not the injury is considered product related Product delivery complaints are issues related to either products or services that has been delivered. As an example, those complaints include wrong deliveries, damaged delivery boxes, later deliveries, wrong quantity shipped, etc. as well as ineffective or untimely services. NOTE: All damaged kit box complaints must be handled as a product complaint.
Learning Management System (LMS)	The Learning Management System (LMS) is used to track classroom and online training for employees, partners and customers. The LMS is accessible through Salesforce.
Non-routine servicing (Non-Routine)	Repairs of an unexpected nature, replacement of parts earlier than their normal life expectancy, or identical repairs or replacements of multiple units of a device are not routine servicing
Out of Box Failure (OBF)	Describes a problem that occurs during the installation of a new or refurbished instrument for which the FSE is obliged either to replace one or more spare parts, components, modules and/or conduct additional troubleshooting not required by the applicable installation/ IQOQ procedure in order to successfully complete installation.
Production Identifier (PI)	Refer to 02643

Removal	The physical removal of the device from its point of use to some other location for repair, modification, adjustment, re-labeling, destruction, or inspection.
Routine servicing (Routine)	Means any regularly scheduled maintenance of a device, including the replacement of parts at the end of their normal life expectancy, e.g., calibration, replacement of batteries, and responses to normal wear and tear.
Salesforce Corporate (SCO)	SCO is a cloud-based Customer Relationship Management (CRM) platform
Service	The repair or maintenance of a product to ensure that it conforms to specifications.
Service pFMEA	Service pFMEA (pFMEA) is used to assess risks associated with the servicing of the manufacturing device
Field Service Technical Documentation	Field Service Technical Documentation is used to disseminate information to Field Service personnel. Refer to SOP 01651.
Service Software	Software used by Field Service to document and record the installation, maintenance, troubleshooting of products distributed by Luminex
Third Party	Any service organization that is not a subsidiary of Luminex Corporation
Unique Device Identifier (UDI)	Refer to 02643
Work Order (WO)	A record used within a Case to assign a Field Service Engineer or the Luminex Central Repair Facility (LCRF) a request to perform service, preventative maintenance, or installation/de-installation; potential for correction or removal action. Refer to 00942 SOP.

5 Process Flowchart

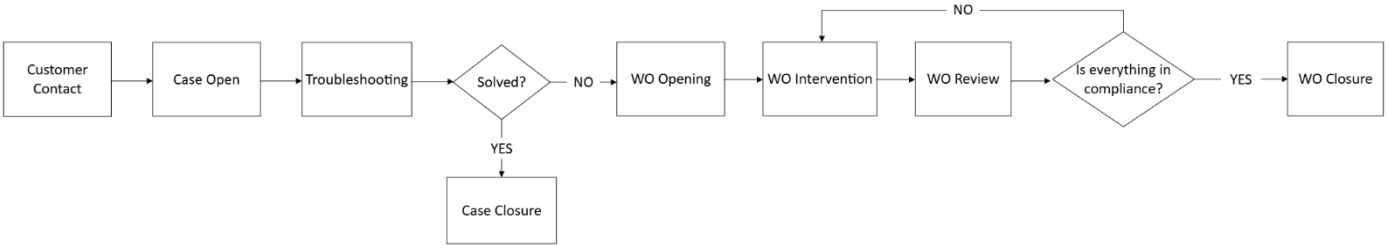


Figure 5-1. Process flowchart

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6 Activities

6.1 Overview of Field Service Activities

- Field Service Activities include and are not limited to the following situations:
 - Installation/Re-Installation based work orders, where the customer requests Field Service engineers to install/re-install instrument systems at the customer site.
 - Routine servicing/Preventive maintenance-based work orders as directed by product specific service manuals
 - Customer complaints that are associated with a priority level 2 (or higher) where field service personnel are dispatched to investigate, troubleshoot and/or repair an instrument in the field.

Refer to 00942, Inquiry and Complaint Handling Operating Procedure, for further details.

 - When repair/service cannot be completed on site, the material is removed via a Return Material Authorization (refer to 03670).
- Work Order is the record used to register all the detailed information performed by the FSE during servicing of an Installed Product at Customer Sites and Internal Labs. The information enclosed in a WO represents a Service Record that is used to demonstrate compliance of Service Operation to both Regulations and QMS requirements.
- Work Orders are opened in the designated Service Software, described in 02883 QMS Software Master List, and they can be managed at two different levels:
 - Serialized chassis, the serial number of a complete system (chassis and modules)
 - Serialized subcomponent, the serial number of a specific module installed into a complete system

6.2 Field Service Training

- Refer to the New Field Service Engineer Onboarding Point of Use Instructions, 03735.
- All Field personal shall be trained, training will include content directed by the Training Operating Procedure, 00091, and will also include a specialized Field Service Training Program.
- The content included in the Field Service training program shall be created per the Field Service Training Content Development SOP, document number 04445.
- Field Service Training is directed by the Creating Field Service Product Training Certification SOP, 04437, and also includes but is not limited to the following topics:
 - On-the-job training
 - Field Service Manuals
 - Field Service Technical Documentation
 - Technical training on instrument
 - Safety
 - ESD

6.3 Field Service Technical Documentation:

- Certified Field Service Engineers will receive product-specific technical documentation, including Technical Notes and service-related content. Any service-related documentation requiring training is available through the internal Learning Management System (LMS).
- Trainees must complete all required documentation within their documentation learning plans before performing field activities or within 30 days of their training assignment, whichever occurs first.

6.4 Field Service Manuals

- Field Service Manuals shall be established and shall outline and maintain instructions and procedures for performing and verifying installation and service activities.

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- Refer to the Field Service Training Content Creation SOP, 04445, and the Building Manuals and Package Inserts Standard Operating Procedure, 02151, for further details on specific content that must be included in manuals.

6.5 Service Product Information

- Service Product information includes and is not limited to the following items:
 - Product specific Field Service Manuals
 - Service-Related Technical Documentation
 - Training Material (Manuals, Handouts, Guides)
 - End User Manuals

6.6 Video and Audio Communication Tools

Field Service may leverage validated and approved remote collaboration tools and software to enhance service support. This includes the use of technologies such as augmented reality, real-time video communication, and other remote assistance solutions that facilitate effective troubleshooting and service delivery.

6.7 Field Service Equipment

- Equipment used during Field Service activity must be maintained according to its calibration schedule as governed by the Calibration and Preventative Maintenance OP, 03003.
- Equipment used in Field Service shall be shipped, handled, stored, and transported as specified in the Inspection, Measuring and Test Equipment SOP (00093), Field Service manuals, and Field Service Technical Documentation.
- Equipment shall be returned from the field according to the process described in 01199, Field Service Parts Return and Management Standard Operating Procedure.

6.8 Field Service Records

- All field service activities shall be documented as Field Service Records in the Luminex approved software system. Refer to 02883, QMS Software Master List.
 - The software system used to fulfill all Service Record requirements is validated per Corp-GOP-000104 , “Non-Product” software application - implementation and validation - handling of Corporate projects and aligns with the respective Field Service Manual.
- Field Service activities are scheduled and managed within the work order management system according to 04080, Field Service Work Order Management SOP.
- Field Service reports shall minimally include:
 - Work Order or equivalent service report identification number
 - Date of Service
 - Customer Information
 - Associated Case # (not applicable to PM work orders)
 - Name of the device serviced
 - Instrument Information
 - o Product identification (e.g., serial/control number)
 - o Unique Device Identifier (UDI DI) - Mandatory for IVD product Only
 - Individual performing service (FSE)
 - Service performed and details
 - Parts Used/Replaced
 - o This includes any reagents used for verification and/or calibration
 - Testing and Inspection Data (Verification and/or calibration data/results)
 - Identification and calibration due date, where applicable, of the reference equipment used

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- If reference equipment can be used for different service activities, such as used for both calibration and verification, the equipment used will be identified for each activity

6.9 Hard copy Field Service Report (All Systems)

In cases where the online work order management system is not available, a hard copy Field Service Report can be utilized to capture service activities utilizing FS-1713-01. All data required to be captured in this procedure will need to be retained and loaded into the system once the online work order management system is available.

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6.10 Routine vs Non Routine

- All routine vs non routine assessments will be managed in SCO;
- Field Service Records will also include the outcome of the assessment for Routine Vs Non Routine according to the flow chart below:

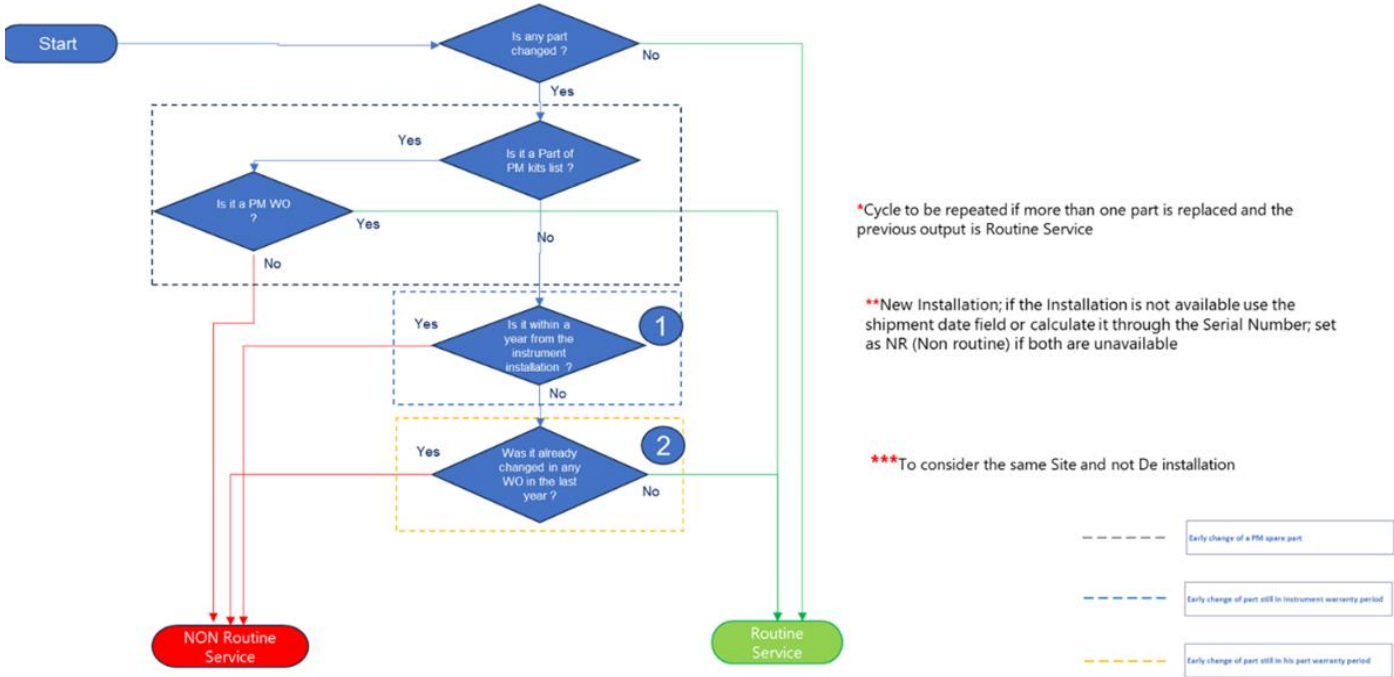


Figure 6-1. Routine vs Non-Routine Assessment Flowchart

- If the part was replaced within a year from the instrument installation (Number '1' in Figure 6-1), reference the flowchart below:

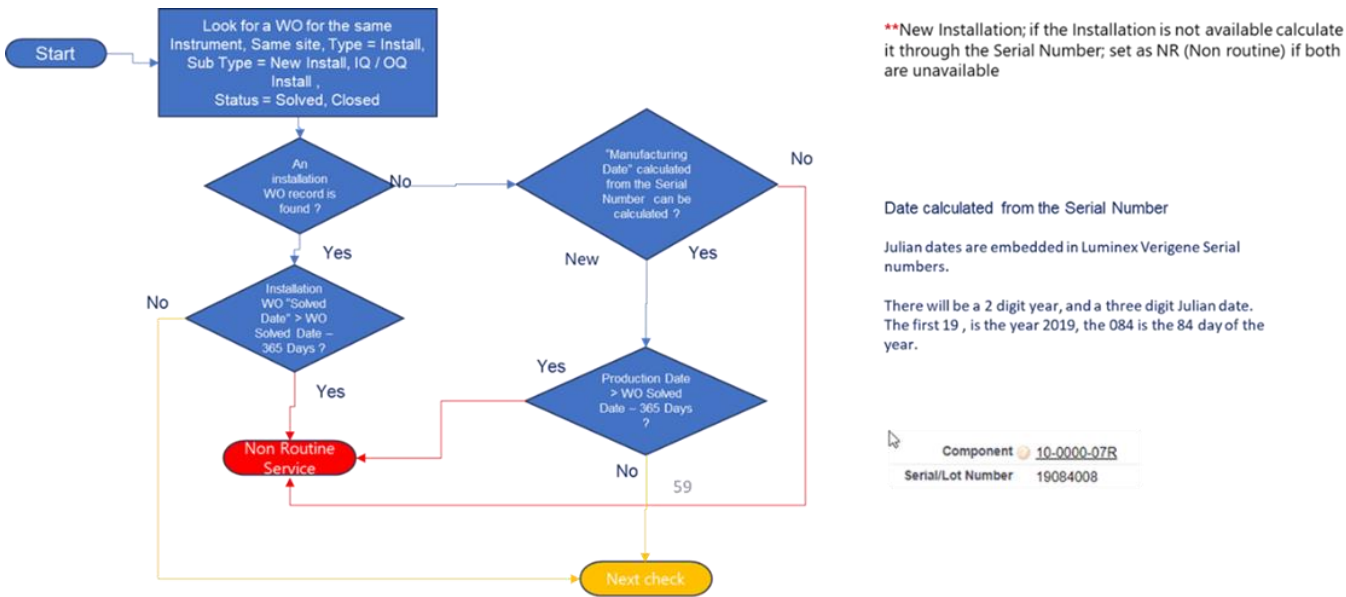


Figure 6-2. Routine vs. Non-Routine Flowchart

- If the part was replaced less than one year after the previous replacement (see Number 2 in Figure 6-1), refer to the flowchart below:

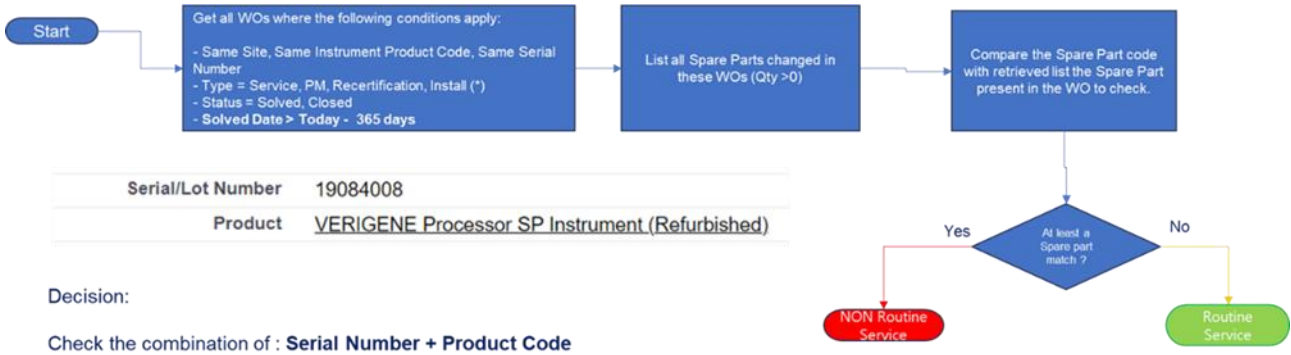


Figure 6-3. Routine vs Non-Routine flowchart

- Results shall be documented per the following criteria:

Service Action Performed	Type	Necessary Action
Preventive Maintenance or Planned Maintenance /install/re-install	<i>Routine</i>	Refer to 00942 for details on documentation requirements within the complaint case.
	<i>Non-Routine</i>	Priority level of case gets upgraded to a Level 3 - Refer to 00942 for further actions.
Complaint/Service Based Work orders	<i>Routine</i>	Refer to 00942 for details on documentation requirements within the complaint case.
	<i>Non-Routine</i>	Complaint case shall be routed to the Field Quality department to perform a work order assessment; refer to 00942 for further details.
	<i>Servicing/troubleshooting or repair activities CANNOT be performed onsite and instrument was returned to Luminex</i>	Refer to 03670 for more information.

Table 1. Field Service Report Analysis Guidelines

6.11 Instrument Configuration Management

- Work Orders can have interventions on module:
 - Work order will show all the serialized subcomponent interventions that have been documented within the work order under the ‘Intervention on Modules’ section of the Work Order.

Instrument
Appointment
Parts & Tools
Checklist
Interventions On Modules
More

Intervention on Modules (2)

2 items • Updated a few seconds ago

	Work Order Num...	Work Type Name	Intervention on Modules	Installed Instrument	
1	01186034	INSTALLATION	Installation	LXSD22102623	
2	01186033	DE-INSTALLATION	De-Installation	LXSD22102623	

View All

6.12 Field Service Activities and Third-Party Service

- Field Service activities are conducted for both molecular and LTG distributed systems and can also apply to Third Party service where Luminex has been contracted to conduct service activities.
- When Field Service Engineers become aware of a potential complaint, they will report it to Technical Support, per the Complaint (Communication) Handling Operating Procedure, 00166.
 - Table 2 outlines service guidelines for Luminex distributed systems

If the service driver is...	Then...	
An installation	- Schedule installation - Perform installation and verification as per Field Service Manual - Document installation information on Field Service Report - Document any nonconformances (OBF) and/or repairs on Field Service Report and complete task assignment notes in the complaint handling system	
A Customer Service Complaint	- Evaluate Technical Support Complaint - Contact the customer - Resolve complaint either verbally or schedule Service Call on site or arrange for unit return and/or replacement	
	If...	Then...
	Problem Resolved Verbally	Complete task assignment notes in the compliant handling system
	Requires on site Field Service Call	- Schedule appointment - Perform on-site troubleshooting and resolve issue. - Verify repair per Field service Manual - Complete Field Service Report – refer to POU 03756 - Document Field Service Report (Refer to Table 1) - Complete task assignment notes in the compliant handling system - Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).
	Subsequent On-Site Service Calls	- Perform on-site troubleshooting and resolve issue - Verify repair per Field Service Manual - Complete Field Service Report – refer to POU 03756 - Document Field Service Report (Refer to Table 1) - Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).
	Requires return for repair or replacement	- Arrange for instrument replacement, if applicable - Arrange for instrument return per SOP 03670 - Complete Field Service Report – refer to POU 03756 - Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).

If the service driver is...	Then...	
Preventative Maintenance	◆ Schedule Preventative Maintenance ◆ Perform Preventative Maintenance and verification as per Field Service Manual ◆ Document Preventative Maintenance in Field Service Report ◆ Document any nonconformances and/or repairs on Field Service Report – refer to POU 03756 ◆ Document Field Service Report (Refer to Table 1) ◆ Complete task assignment notes in the compliant handling system ◆ Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).	
Recertification	◆ Schedule Recertification ◆ Evaluate equipment and notify customer of all parts and the associated cost that will be required for recertification. Clarify if PO or estimate from management is needed.	
	If...	Then...
	Customer agrees to the costs	◆ Perform recertification and verification per the Field Service Manual ◆ Document recertification information on Field Service Report, refer to POU 03756 ◆ Complete task assignment notes in compliant handling system ◆ Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).
	Customer does not agree to costs	◆ Document information on Field Service Report, refer to POU 03756 ◆ Complete task assignment notes in the compliant handling system ◆ Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).
Hardware or Software upgrade	◆ Schedule site visit ◆ Perform upgrade and requalification tests as per instruction in the Field Service technical documentation ◆ Document upgrade on Field Service Report ◆ Complete Work Order Review and Closure following POU 03784 which includes risk classification for Service Activities according to Service pFMEA (service pFMEA is available for each instrument line and requirements are reflected in POU 03784 attachments).	

Table 2. Service Guidelines for Luminex Distributed systems and providing Third Party Service

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6.13 Service Activities subcontracted to Third-Party Entity

In situations where the company has decided to subcontract out service activities to a third-party entity the following shall be required:

- The third party shall be identified, classified and qualified according to the Corporate Distribution Programs Operating Procedure (02381) and the Supplier Qualification and Management Program Standard Operating Procedure (01836).
- In addition to qualification, this type of relationship shall require a Quality Agreement be put in place with the third party as directed by the Legal Contract Standard Operating Procedure 00378 and the Management Program Standard Operating Procedure, 01836.

6.14 Field Service Parts

- Field Service Engineers are responsible for maintaining adequate levels of trunk stock according to the Field Service Inventory Management SOP, 03971.
- Field Service Engineer spare parts should be ordered according to the Parts Ordering Point of Use Instructions, 03999.
- Field Service person can return parts from system to WEEE recycling returns per the Returning Defective Inventory To Manufacturing Standard Operating Procedure, 00935.

7 Local Requirements

This procedure covers the basics needed to perform the described activity. Local procedures may not be necessary.

8 Responsibilities

Role	Responsibilities
Global Service Support Specialists	◆ Create Field Service manuals, Field Service Technical Documentation, and training material ◆ Provide services, such as repair or preventive maintenance, as needed. ◆ Transmit product information to Internal and External Field Service Engineers. ◆ Analyze field service records per 03491, Global Service Operating Procedure, for continual improvement and/or trending. Evaluate the need for additional investigation per 00283, Corrective and Preventive Action Standard Operating Procedure.
Field Service & Third Party	◆ Receive updates from the Global Service Support Specialists ◆ Document service activities and maintain records ◆ Document Field Servicer records for routine and non-routine based activities
Field Quality	◆ Perform work order assessments for non-routine based activities from work orders. ◆ Refer to 03805, Field Quality Work Order Assessment Point of Use Instructions
Technical Support	◆ Analyze complaints associated with field service dispatch for reportable events and notifies Regulatory Affairs of injury reports or PRE situations. ◆ Close customer complaints, as applicable
Service Operations Support	◆ Manage Returned Materials Authorization (RMA) process, as primary ◆ Manage install base, install, deinstallations and PM records, as primary, mainly within North America ◆ Shipping and Logistics related complaints (i.e. outside container) ◆ Use and create Articles in the Knowledge Base for uniformity in handling issues ◆ Provide complaint data analysis for Quality/Management Review and CAPA consideration ◆ Can fulfill Technical Support duties as a backup ◆ Refer to 00942, Global Support Services Communication Standard Operating Procedure

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